# Activity 5

## Todo: #1Create Database

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "CREATE TABLE authors (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    country VARCHAR(50)
);"
Password for user aditya:
CREATE TABLE
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "CREATE TABLE books (
    id SERIAL PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author_id INTEGER REFERENCES authors(id),
    price DECIMAL(10,2),
    published_year INTEGER
);"
Password for user aditya:
CREATE TABLE
aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #2 Create Tables

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "CREATE TABLE authors (
    id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    country VARCHAR(50)
);"
Password for user aditya:
CREATE TABLE
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "CREATE TABLE books (
    id SERIAL PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    author_id INTEGER REFERENCES authors(id),
    price DECIMAL(10,2),
    published_year INTEGER
);"
Password for user aditya:
CREATE TABLE
aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #3-4  Insert Data into tables

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "INSERT INTO authors (name, country) VAL
UES
    ('Margaret Atwood', 'Canada'),
    ('George Orwell', 'United Kingdom'),
    ('Haruki Murakami', 'Japan');"
Password for user aditya:
INSERT 0 3
aditya@morteza-s-S24-Ultra ~ %
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "INSERT INTO books (title, author_id, pr
ice, published_year) VALUES
    ('The Handmaid''s Tale', 1, 15.99, 1985),
    ('1984', 2, 12.99, 1949),
    ('Animal Farm', 2, 10.99, 1945),
    ('Norwegian Wood', 3, 14.99, 1987);"
Password for user aditya:
INSERT 0 4
aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #5 Select Query with condition

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "SELECT * FROM books WHERE price < 14.00
;";
Password for user aditya:
 id |     title     | author_id | price | published_year
----+---------------+-----------+-------+----------------
  2 | 1984          |         2 | 12.99 |           1949
  3 | Animal Farm   |         2 | 10.99 |           1945
(2 rows)

aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #6 Select Query with join

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "SELECT books.title, authors.name AS aut
hor, books.price
FROM books
LEFT JOIN authors ON books.author_id = authors.id;";
Password for user aditya:
        title        |     author      | price
---------------------+-----------------+-------
 The Handmaid's Tale | Margaret Atwood | 15.99
 1984                | George Orwell   | 12.99
 Animal Farm         | George Orwell   | 10.99
 Norwegian Wood      | Haruki Murakami | 14.99
(4 rows)

aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #7 update query

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "UPDATE books SET price = 13.99 WHERE ti
tle = '1984';";
Password for user aditya:
UPDATE 1
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "SELECT * FROM books WHERE title='1984';
";
Password for user aditya:
 id | title | author_id | price | published_year
----+-------+-----------+-------+----------------
  2 | 1984  |         2 | 13.99 |           1949
(1 row)

aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #8 Delete Query

```
aditya — -zsh — 91×23
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "DELETE FROM books WHERE title = 'Animal
 Farm';";
Password for user aditya:
DELETE 1
aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "SELECT * FROM books;";
Password for user aditya:
 id |        title        | author_id | price | published_year
----+---------------------+-----------+-------+----------------
  1 | The Handmaid's Tale |         1 | 15.99 |           1985
  4 | Norwegian Wood      |         3 | 14.99 |           1987
  2 | 1984                |         2 | 13.99 |           1949
(3 rows)

aditya@morteza-s-S24-Ultra ~ %
```

## Todo: #Bonus using aggregate functions

```
aditya — -zsh — 91×23
JOIN authors ON books.author_id = authors.id
GROUP BY authors.name
HAVING COUNT(*) > 1;";
Password for user aditya:
 name | average_price | book_count
------+---------------+------------
(0 rows)

aditya@morteza-s-S24-Ultra ~ % psql -d postgres -c "SELECT authors.name, AVG(books.price) A
S average_price, COUNT(*) AS book_count
FROM books
JOIN authors ON books.author_id = authors.id
GROUP BY authors.name
HAVING COUNT(*) = 1;";
Password for user aditya:
      name       |     average_price     | book_count
-----------------+-----------------------+------------
 George Orwell   | 13.9900000000000000 |          1
 Margaret Atwood | 15.9900000000000000 |          1
 Haruki Murakami | 14.9900000000000000 |          1
(3 rows)

aditya@morteza-s-S24-Ultra ~ %
```